ROVER ARCHITECTURE & OBSTACLE-AVOIDANCE ALGORITHM

Obstacle-Avoider Design Challenge • UG2–UG4 Elimination Round

Team: WattNots | **Members:** Anantharaman VL, Navajyoth, Fijo

1. Rover Architecture

WattNots' rover uses a two-wheel differential-drive architecture. Two DC geared motors independently drive the left and right wheels, while a caster wheel provides support. An Arduino Uno acts as the main controller, receiving distance measurements from an HC-SR04 ultrasonic sensor and commanding the motors through an **L298N dual H-bridge motor-driver module**. A servo is included to point the ultrasonic sensor toward left, centre and right scan positions. The battery supplies the motor system, while the controller and sensor electronics use their required logic supply; all grounds share a common reference.

The chassis was designed and assembled by the team in Autodesk Fusion 360. The custom structure includes the chassis decks, motor brackets, structural pillars, sensor/servo mounting arrangement, and provisions for the battery and electronics/PCB. Standard wheels, motors and sensors are used as components; the rest of the structure was designed by the team rather than copied from an existing chassis.

2. Component Choices

Component	Role and selection rationale
Arduino Uno R3	Runs sensing, decision-making and motor-control code; provides accessible I/O for the prototype.
HC-SR04 ultrasonic sensor	Measures obstacle distance using an ultrasonic pulse and echo.
SG90-compatible servo	Moves the sensor between left, centre and right positions for the intended scan strategy.
L298N dual H-bridge motor-driver module	The selected driver for the intended rover design. It controls the direction of both DC motors. The Tinkercad circuit uses an L293D/L298D-labelled substitute only because a suitable L298N model was unavailable in the simulator; this does not change the intended physical driver.
Two DC geared motors and drive wheels	Provide differential steering; different motion on the left and right sides turns the rover.
Caster wheel	Provides a third support point for the two-wheel-drive chassis, if included in the final assembly.
Battery and power wiring	Supply the motor driver and controller/sensor system. Physical battery, protection and regulation must match component ratings.

3. Obstacle-Avoidance Algorithm

The firmware repeatedly measures the front distance and compares it with a configured safety threshold. When the path is clear, the rover moves forward. When an obstacle is detected within the threshold, it stops and commands the servo to scan left and right. The readings are compared and a side that appears clearer is selected. The rover performs a timed turn, advances to try to pass the obstacle, and checks the front distance again. If both scan directions appear blocked, it reverses briefly before trying again.

Step	Stage	Action
1	Sense	Read the HC-SR04 distance at the forward/centre position.
2	Decide	If distance exceeds the configured threshold, continue forward; otherwise stop and initiate avoidance.
3	Scan	Move the servo to left and right scan angles and record readings.
4	Choose	Select a direction that appears clear; if neither side appears clear, reverse briefly.
5	Avoid	Turn using differential motor control, then advance for a tuned interval.
6	Recheck	Measure front distance again and resume forward movement when considered clear.

4. Simulation Scope and Limitations

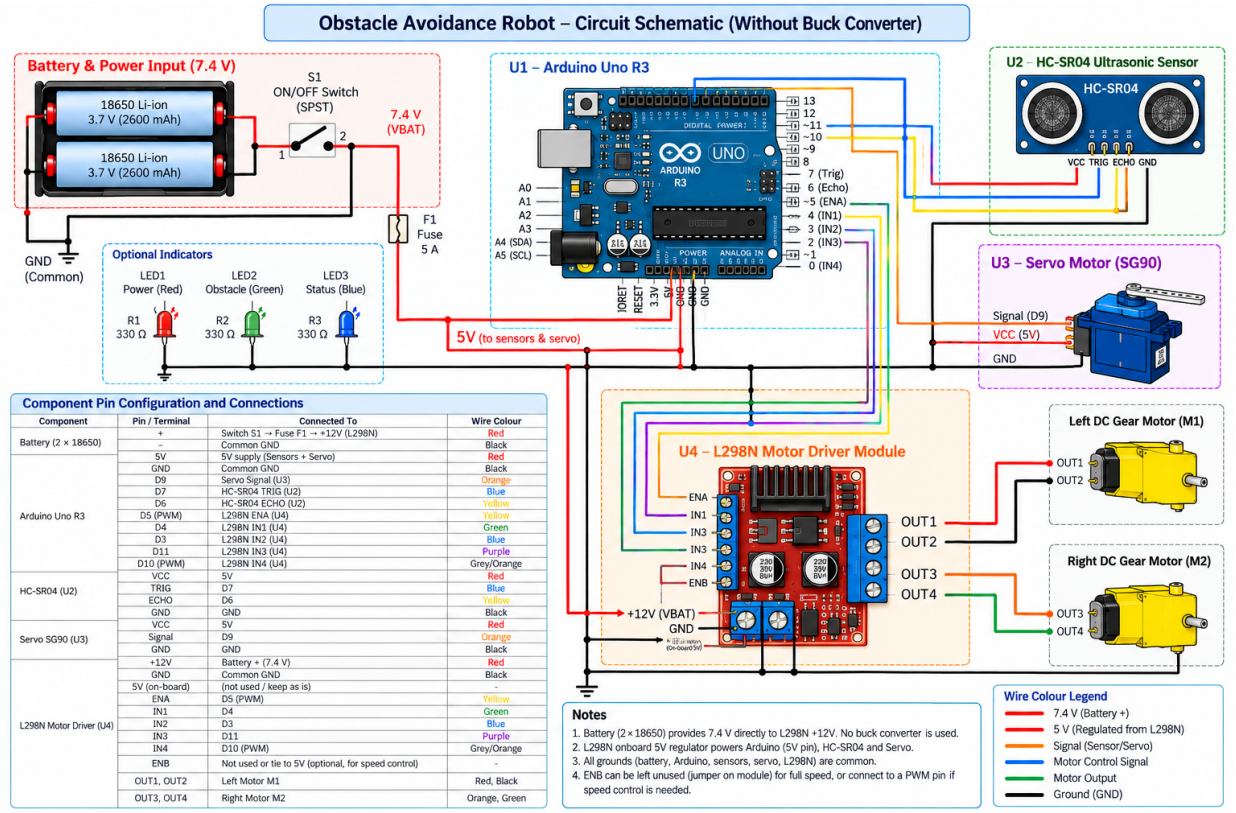
The project is being validated through circuit simulation and CAD modelling; no manufactured PCB or completed physical rover is claimed.

The L298N is the specified motor driver for the intended design and physical BOM. The L293D/L298D-labelled component is used only as a Tinkercad simulation substitute because a suitable L298N model could not be found. The final physical-design schematic and BOM should specify L298N. In Tinkercad, the servo and HC-SR04 are not mechanically linked, so servo movement may not change the sensor's field of view. Timed turns and forward motion are approximate because wheel encoders are not used to measure distance or heading.

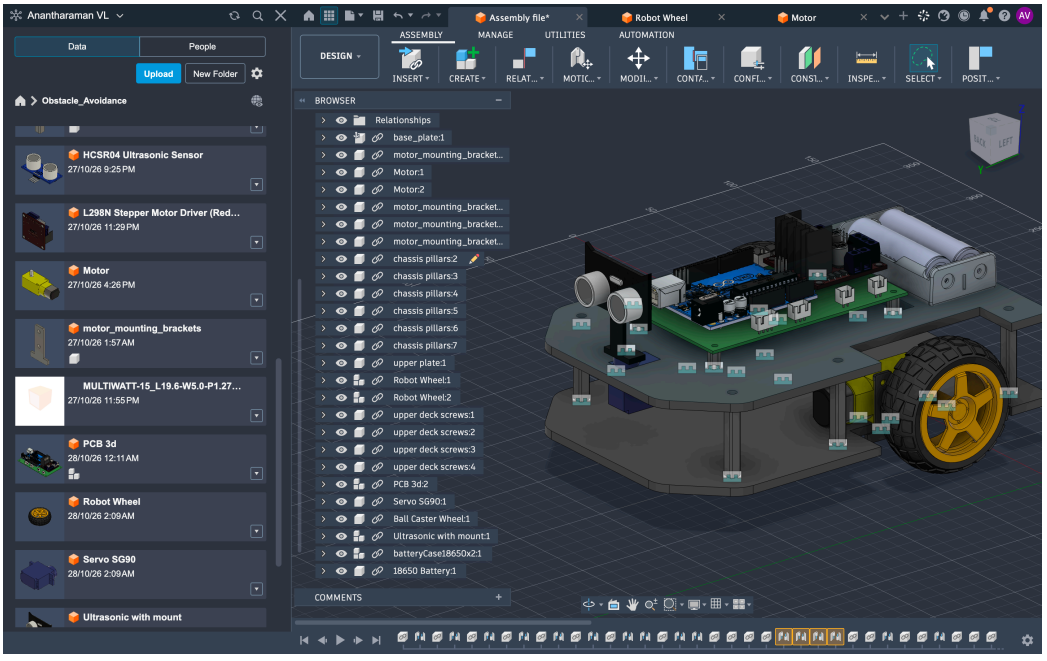
Required behaviours to validate

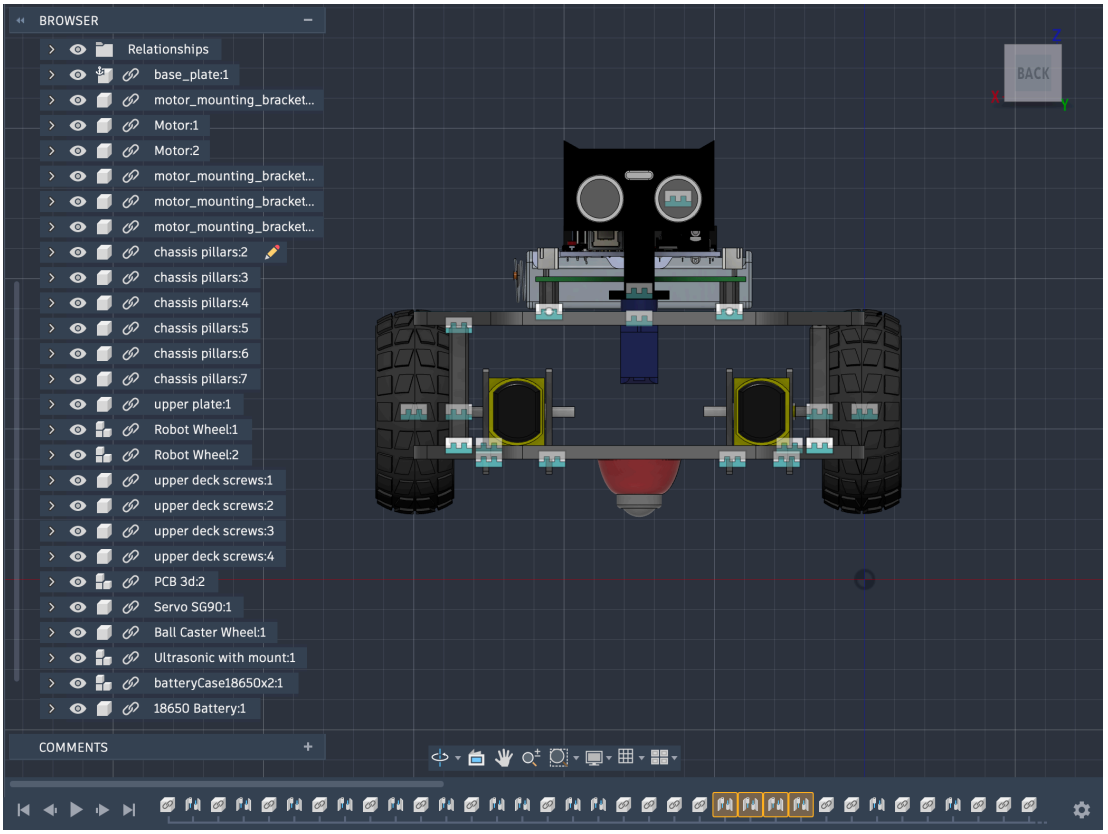
- No obstacle within the threshold: move forward.
- Obstacle ahead: stop or change movement to avoid collision.
- Avoidance manoeuvre: turn or adjust direction.
- Path clear: resume forward movement when safe.

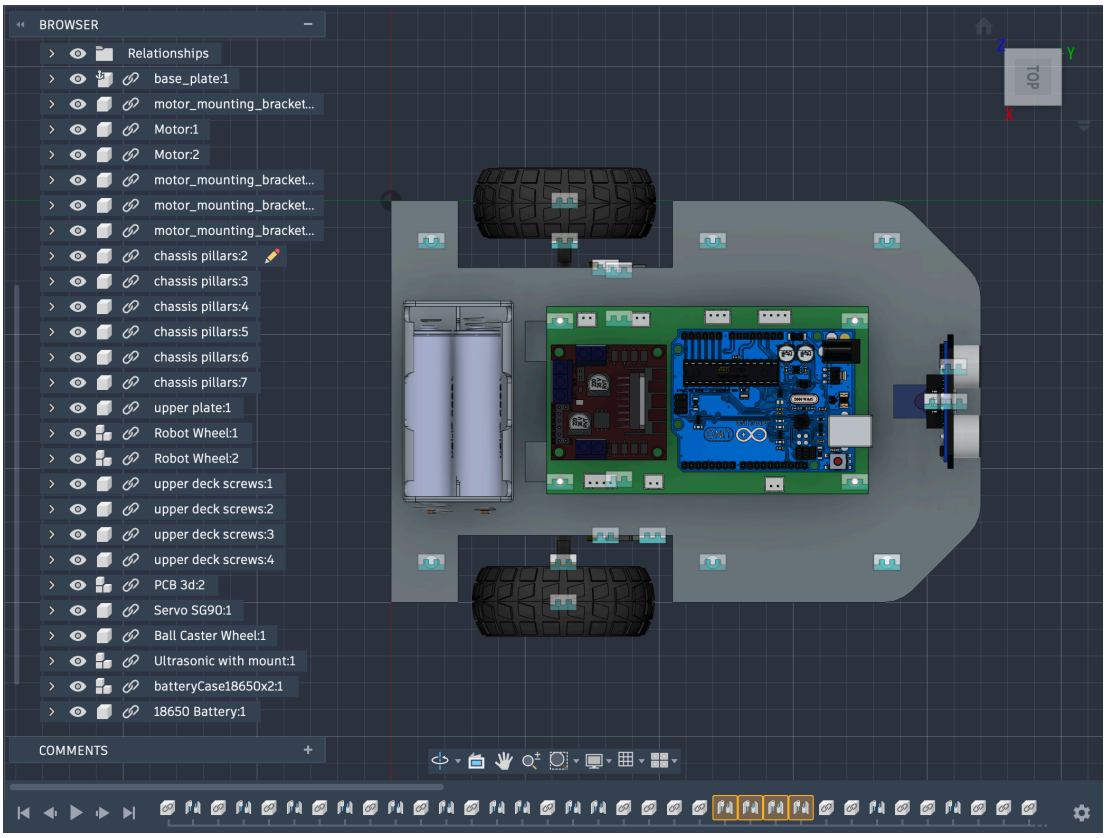
System Architecture

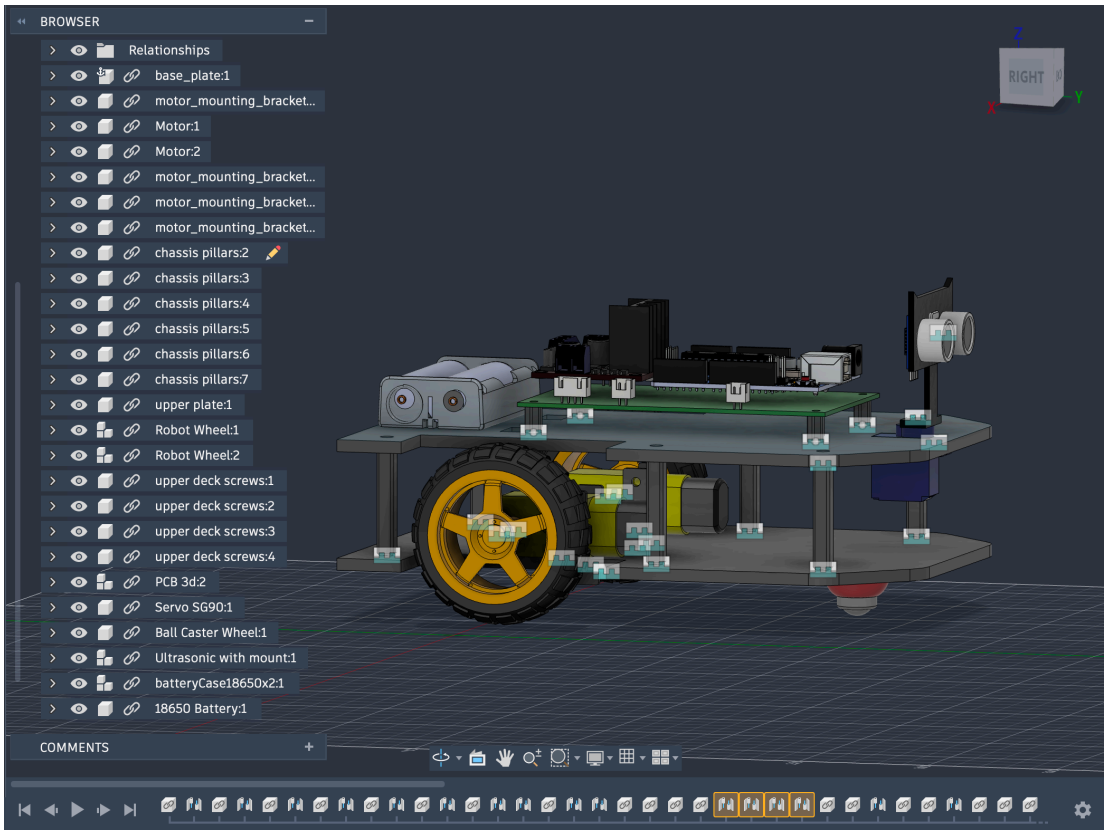


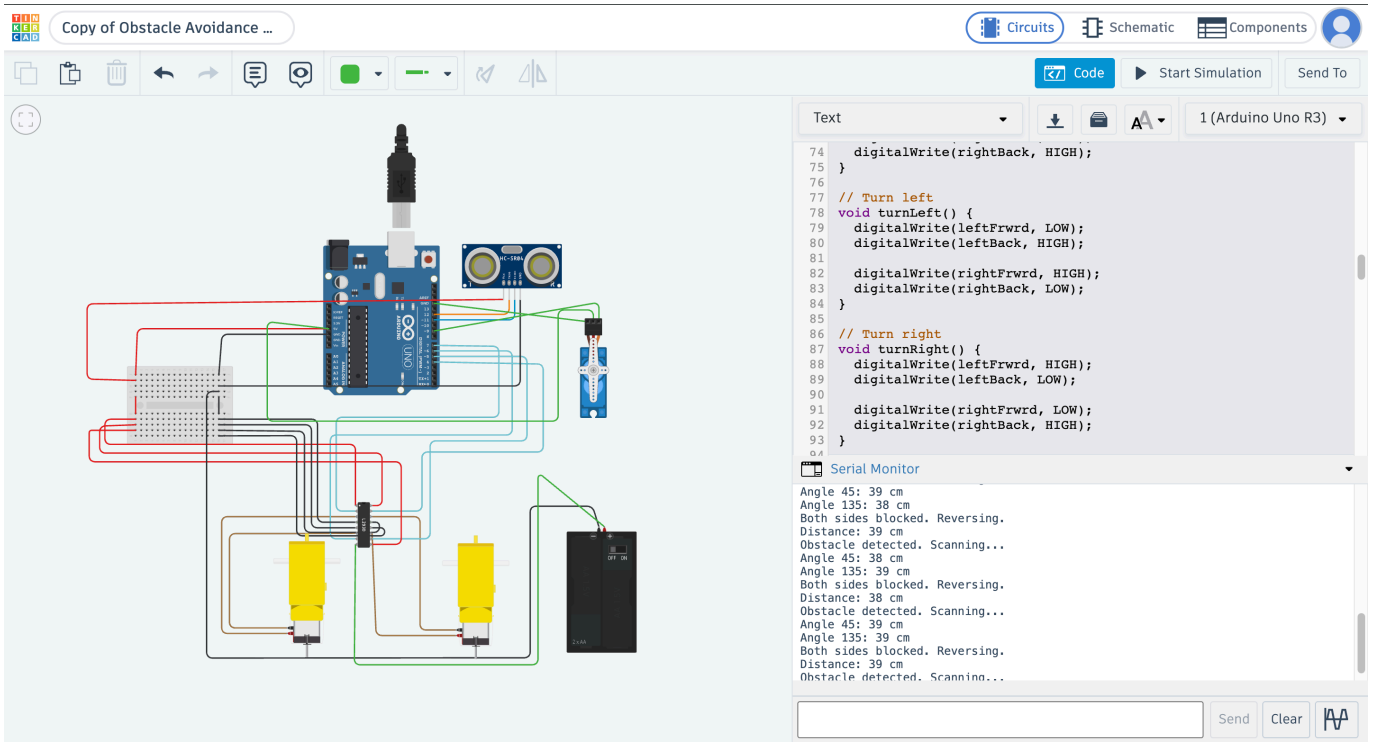
ScreenShots and Obseravtions

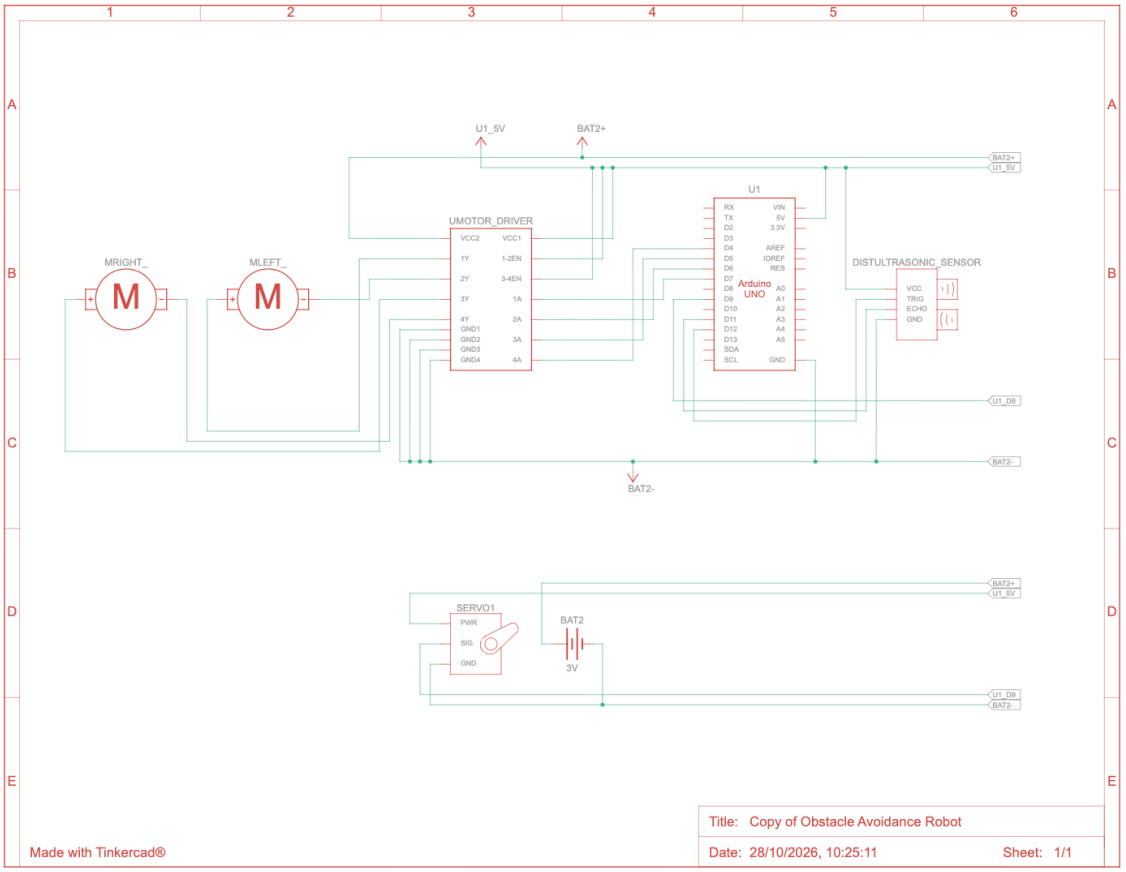



BROWSER
Relationships
base_plate:1
motor_mounting_bracket...
Motor:1
Motor:2
motor_mounting_bracket...
motor_mounting_bracket...
motor_mounting_bracket...
chassis pillars:2
chassis pillars:3
chassis pillars:4
chassis pillars:5
chassis pillars:6
chassis pillars:7
upper plate:1
Robot Wheel:1
Robot Wheel:2
upper deck screws:1
upper deck screws:2
upper deck screws:3
upper deck screws:4
PCB 3d:2
Servo SG90:1
Ball Caster Wheel:1
Ultrasonic with mount:1
batteryCase18650x2:1
18650 Battery:1
COMMENTS
BACK


BROWSER
Relationships
base_plate:1
motor_mounting_bracket...
Motor:1
Motor:2
motor_mounting_bracket...
motor_mounting_bracket...
motor_mounting_bracket...
chassis pillars:2
chassis pillars:3
chassis pillars:4
chassis pillars:5
chassis pillars:6
chassis pillars:7
upper plate:1
Robot Wheel:1
Robot Wheel:2
upper deck screws:1
upper deck screws:2
upper deck screws:3
upper deck screws:4
PCB 3d:2
Servo SG90:1
Ball Caster Wheel:1
Ultrasonic with mount:1
batteryCase18650x2:1
18650 Battery:1
COMMENTS
TOP


BROWSER
Relationships
base_plate:1
motor_mounting_bracket...
Motor:1
Motor:2
motor_mounting_bracket...
motor_mounting_bracket...
motor_mounting_bracket...
chassis pillars:2
chassis pillars:3
chassis pillars:4
chassis pillars:5
chassis pillars:6
chassis pillars:7
upper plate:1
Robot Wheel:1
Robot Wheel:2
upper deck screws:1
upper deck screws:2
upper deck screws:3
upper deck screws:4
PCB 3d:2
Servo SG90:1
Ball Caster Wheel:1
Ultrasonic with mount:1
batteryCase18650x2:1
18650 Battery:1
COMMENTS
RIGHT


Copy of Obstacle Avoidance ...
Circuits
Schematic
Components
Code
Start Simulation
Send To
Text
1 (Arduino Uno R3)
digitalWrite(rightBack, HIGH);
}
// Turn left
void turnLeft() {
digitalWrite(leftFrwrd, LOW);
digitalWrite(leftBack, HIGH);
digitalWrite(rightFrwrd, HIGH);
digitalWrite(rightBack, LOW);
}
// Turn right
void turnRight() {
digitalWrite(leftFrwrd, HIGH);
digitalWrite(leftBack, LOW);
digitalWrite(rightFrwrd, LOW);
digitalWrite(rightBack, HIGH);
}
Serial Monitor
Angle 45: 39 cm
Angle 135: 38 cm
Both sides blocked. Reversing.
Distance: 39 cm
Obstacle detected. Scanning...
Angle 45: 38 cm
Angle 135: 39 cm
Both sides blocked. Reversing.
Distance: 38 cm
Obstacle detected. Scanning...
Angle 45: 39 cm
Angle 135: 39 cm
Both sides blocked. Reversing.
Distance: 39 cm
Send
Clear



Name	Quantity	Component
U1	1	Arduino Uno R3
MLeft MRight	2	Hobby Gearmotor
UMotor Driver	1	H-bridge Motor Driver
DISTultrasonic sensor	1	Ultrasonic Distance Sensor (4-pin)
3	1	Breadboard Mini
SERVO1	1	Positional Micro Servo
Bat2	1	2 batteries, AA, yes 1.5V Battery